



For an Open Mind

UGANDA TECHNOLOGY AND MANAGEMENT UNIVERSITY

UTAMU

DATABASE MANAGEMENT SYSTEMS

GROUP COURSEWORK

ERD ASSIGNMENT

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INTRODUCTION TO ERD NOTATIONS

- 1. Chen's ERD Notation:** Chen's notation is a conceptual ERD notation that represents entities, attributes, and relationships as entities using rectangles, attributes using ovals (primary keys underlined), relationships using diamonds between entities, cardinality written near relationships (e.g., 1:M)
- 2. Crow's Foot ERD Notation:** Crow's Foot notation is a practical ERD notation used for relational database design where entities are represented using rectangles (with attributes inside), and relationships are represented by lines with symbols.

INVENTORY MANAGEMENT SYSTEM IDENTIFIED ENTITIES

1. Warehouse

- warehouse_id (PK)
- location

2. Section (A warehouse contains multiple sections)

- section_id (PK)
- name
- warehouse_id (FK → Warehouse)

3. Category (Products belong to a category)

- category_id (PK)
- name

4. Product (Represents the inventory items)

- product_id (PK)

- name
- description
- stock_quantity
- price
- category_id (FK → Category)
- section_id (FK → Section)
- supplier_id (FK → Supplier)

5. Supplier (Supplier supplies products)

- supplier_id (PK)
- name
- contact_info
- location

6. Purchase Order (Tracks stock purchases)

- purchase_order_id (PK)
- supplier_id (FK → Supplier)
- date
- total_amount

7. Purchase Order Details (Links purchases to products)

- purchase_detail_id (PK)
- purchase_order_id (FK → Purchase Order)
- product_id (FK → Product)
- quantity

8. Customer (Customers makes sales)

- customer_id (PK)
- name
- contact_info

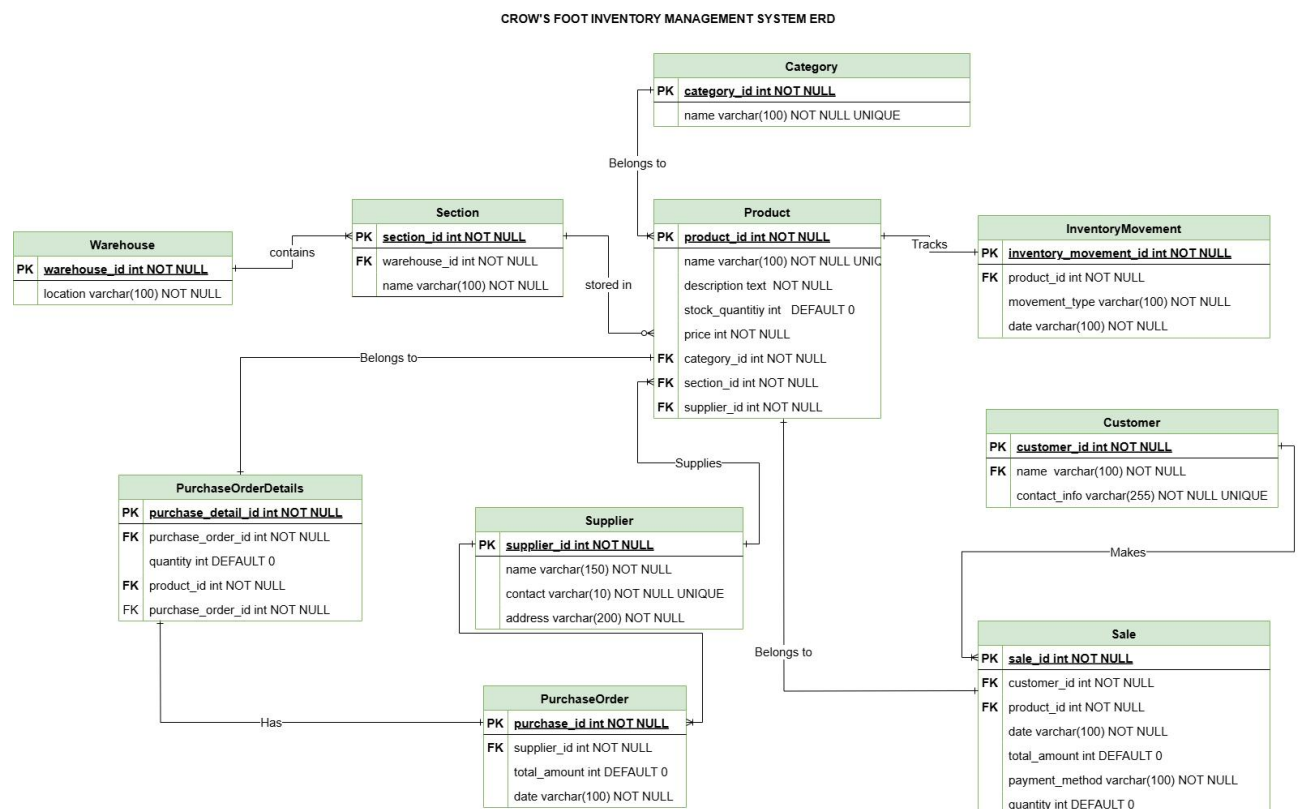
9. Sale (Records customer purchases)

- sale_id (PK)
- customer_id (FK → Customer)
- date
- total_amount
- payment_method
- quantity
- product_id (FK → Product)

10. Inventory Movement (Tracks stock updates)

- movement_id (PK)
- product_id (FK → Product)
- movement_type (in-stock,out-of-stock)
- date

Inventory Management System using Crow's Foot ERD Notation



Inventory Management System using Chen's ERD Notation

CHEN'S INVENTORY MANAGEMENT SYSTEM ERD

